



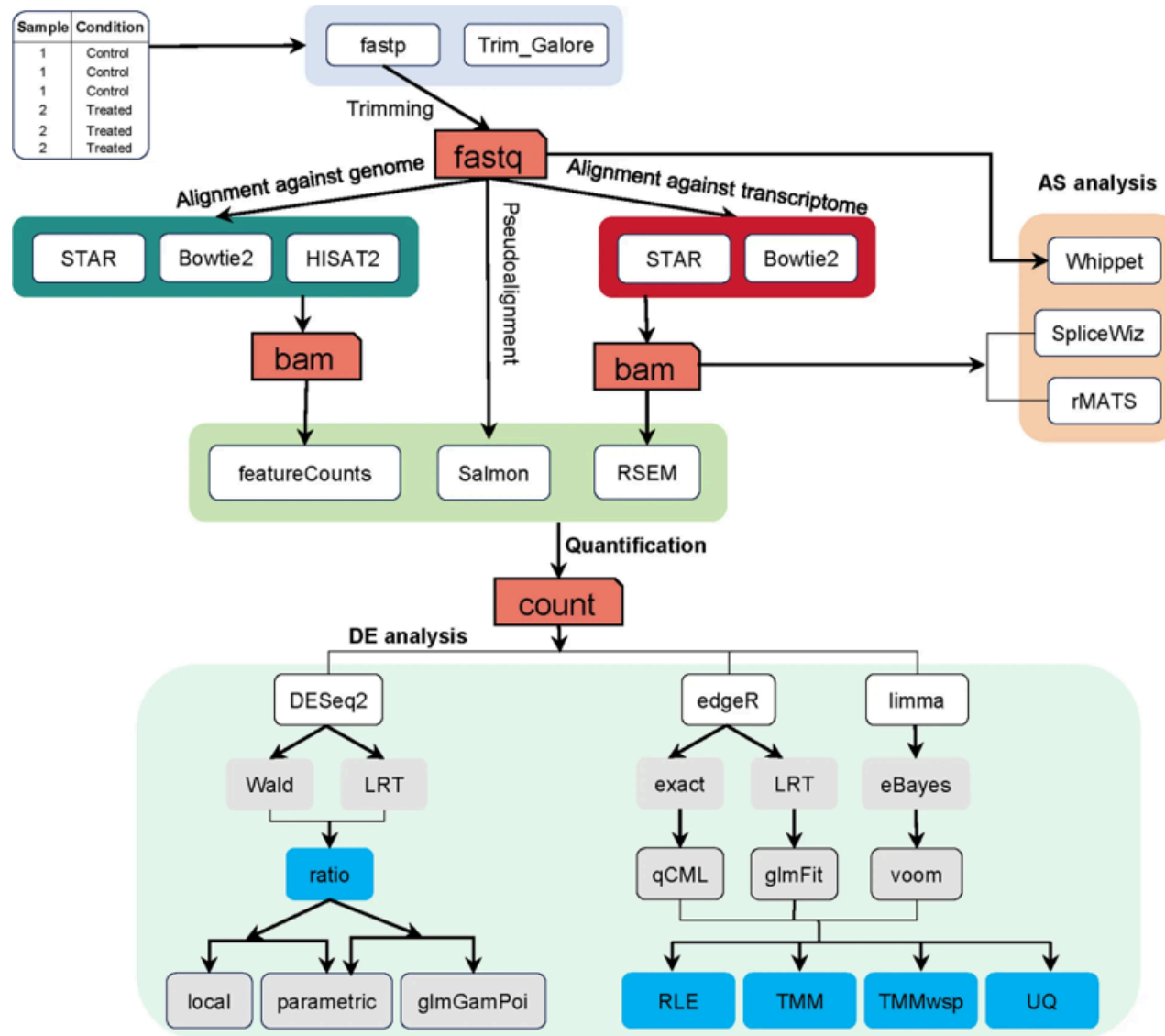
# Learning outcomes

- Overview of transcriptomics and proteomics
- Understand gene coexpression network
- Perform weighted gene coexpression network analysis
- Understand mass spectrometry-based proteomics (MS-Proteomics)
- Perform MS-Proteomics data analysis
- Create a reproducible and publication-ready outputs

# RNA-Seq

- RNA-Seq is the commonly used method for transcriptomic analysis in the era of genomics technologies. This is due to its ability to detect novel transcripts and its high accuracy.
  - It enables the quantification of gene expression levels and identification of genes that exhibit different expression pattern between samples or state in a particular tissue and at a specific time.
  - The goals of RNA-Seq analysis is to identify differentially expressed and co-regulated genes and to infer functional biological implications.
- Previously, micro-array was the go-to techniques for obtaining expression profiles of biological organisms.

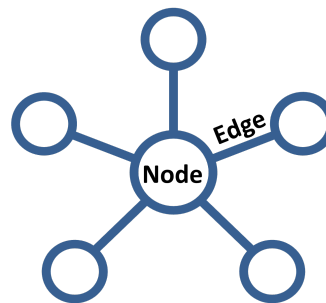
# RNA-Seq Workflow



Jiang et al. 2024 (BMC Genomics)

# RNA-Seq: Gene Coexpression Network

- Gene Coexpression Network (GCN) is an undirected graph for constructing relationships between co-expressed genes. The **node** of a network denotes a particular gene while the **edges** represent co-expression relationship between two genes.
- GCN also helps to identify 'hub gene', i.e, highly connected gene
- Whereas GCN is undirected, gene regulatory network is directed.
- Once gene expression profiles have been normalized from their raw count, GCN analysis (GCNA) can be performed.



Network schematic

## GCN Construction

The network construction entails two steps: co-expression similarity and threshold selection

- Co-expression similarity: Given a normalized gene expression dataframe of  $m$  genes for  $n$  samples, a similarity score is obtained using measures such as Pearson's correlation coefficient. For a chosen measure, the similarity score is obtained between  $i$  and  $j$  genes to form a **similarity matrix**,

$$s_{ij} = r(\mathbf{x}_i, \mathbf{x}_j)$$

- Threshold selection: Given the similarity matrix, a threshold is selected (e.g.  $s_{ij} = |r(\mathbf{x}_i, \mathbf{x}_j)| > 0.8$ ). An element in the similarity matrix above the threshold receives a value 1 otherwise 0. This will result in an **adjacency matrix** where genes connected in a network have 1, else 0.

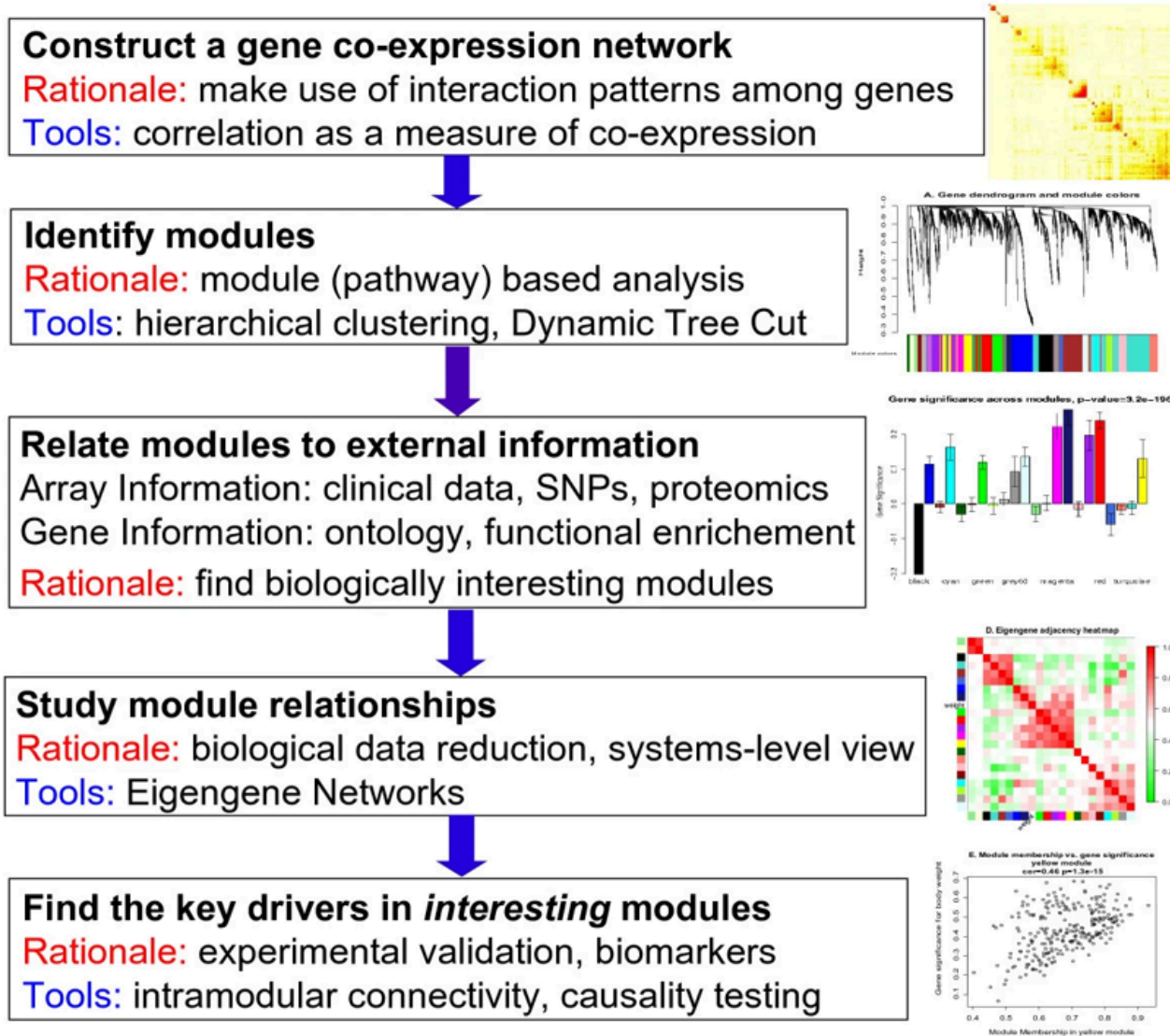
$$a_{ij} = \begin{cases} 1 & \text{for } s_{ij} > 0.8 \\ 0 & \text{elsewhere} \end{cases}$$

# Weighted Gene Coexpression Network Analysis (WGCNA)

The main idea of WGCNA is that weight is assigned to the pairwise relationship between genes (nodes) to obtain the adjacency matrix. That is,  $a_{ij} = s_{ij}^{\beta}$  which implies that,  $\log(a_{ij}) = \beta \times \log(s_{ij})$ , where  $\beta \geq 1$  represents the weight.

- To implement the WGCNA in R, the **WGCNA** package offers useful functions for constructing networks, selecting threshold, and for identifying modules or clusters of genes in a network.

# Overview of WGCNA methodology



Langfelder and Horvath 2008 (BMC Bioinformatics)

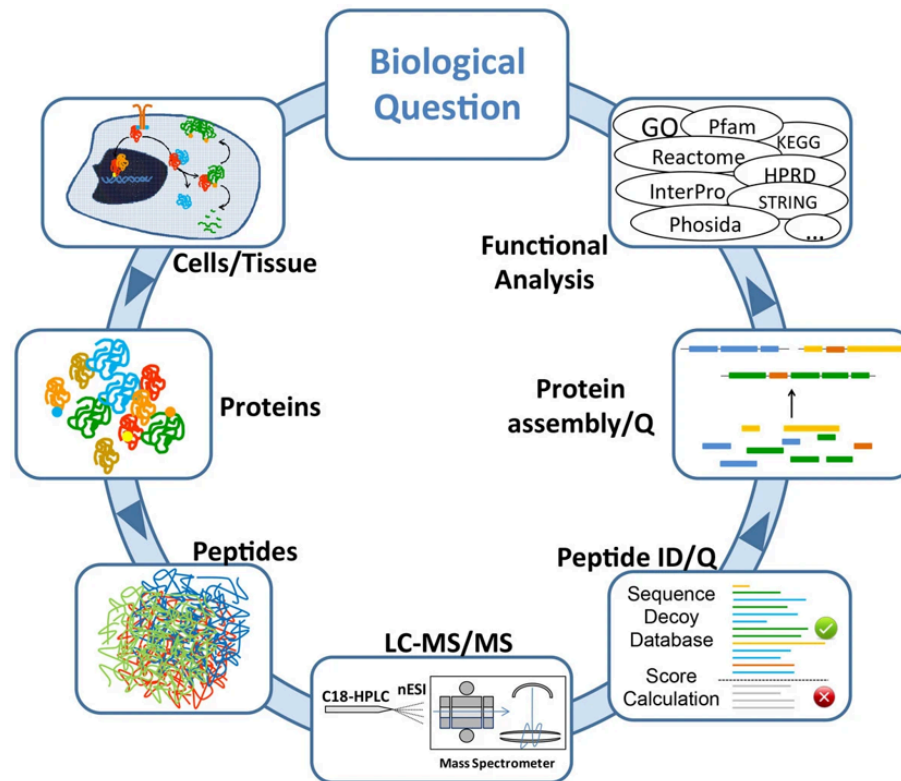




# Proteomics Analysis

Proteomics analysis involves the extraction, fragmentation, quantification, and identification of proteins in a biological sample.

## Proteomics workflow



Schmidt et al 2014 (BMC Syst Biol.)

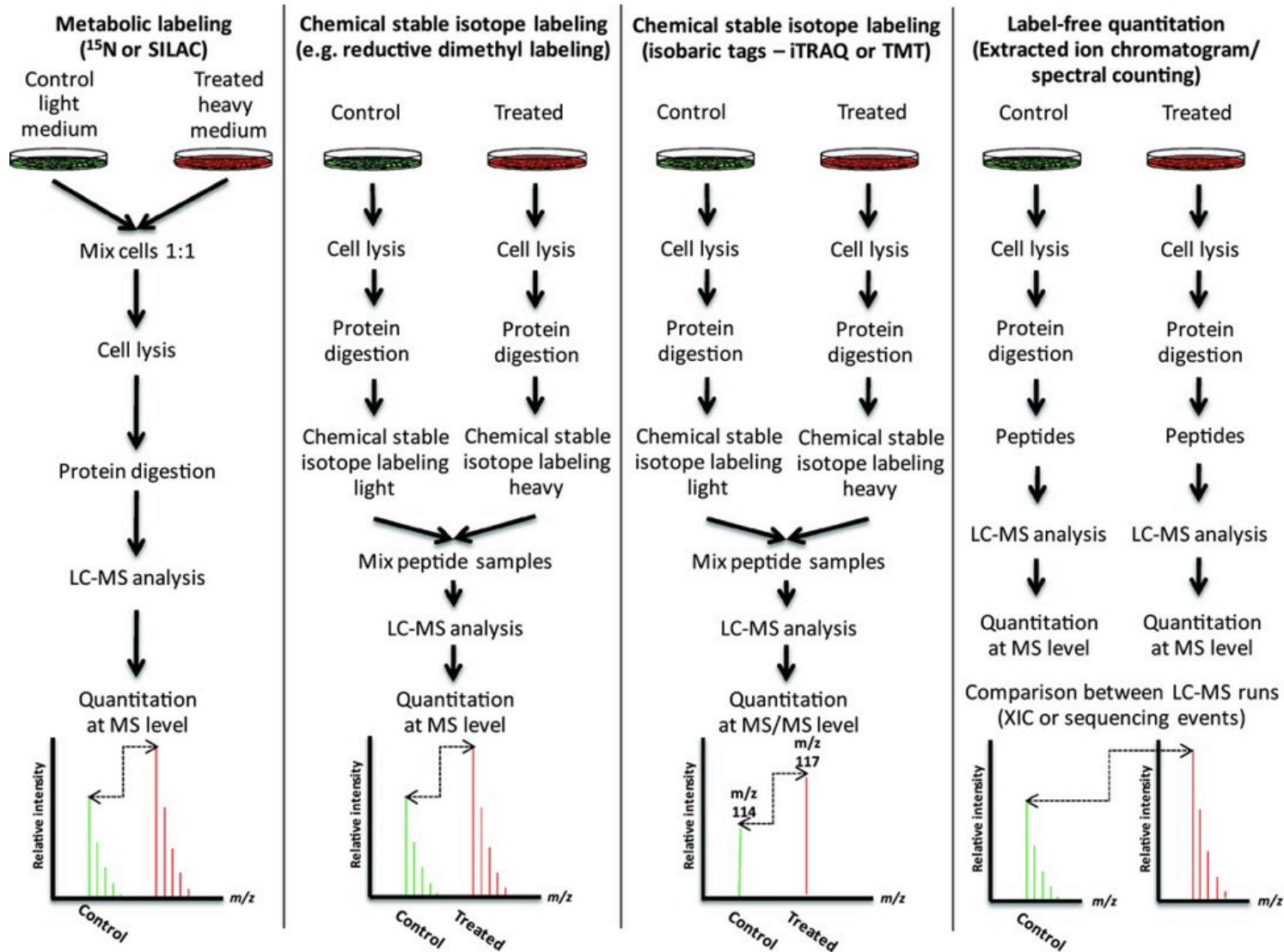
# Quantitative Proteomics

Quantitative proteomics entails the quantification of protein abundance in a biological sample. This is commonly performed using mass spectrometry.

Several methods are being used:

- Targeted
  - Selective ion monitoring (SIM) - Selected reaction monitoring (SRM)
  - Absolute quantitation (AQUA)
- Discovery-based
  - Metabolic labeling: stable isotope labeling by amino acids in cell culture (SLAC),  $^{15}\text{N}$ ,  $^{18}\text{O}$
  - Chemical labeling: isobaric tags for relative and absolute quantification (iTRAQ), tandem-mass tag (TMT), Isotope coded affinity tags (ICAT)
  - Label-free quantification

# Different quantitative proteomics workflow



Engholm-Keller & Larsen 2013 (Proteomics)

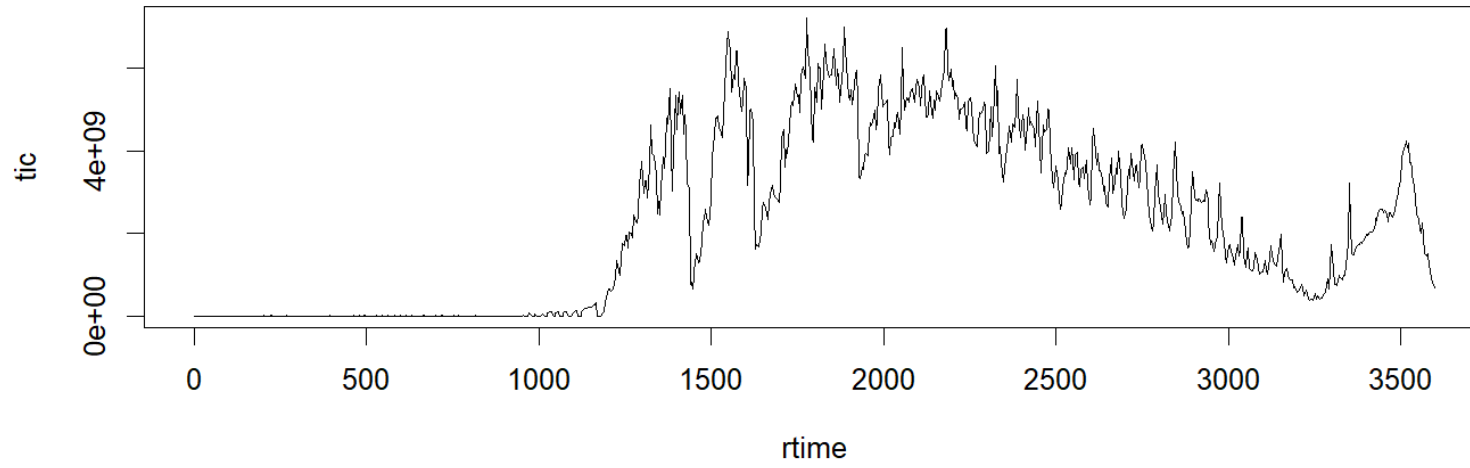
# Brief Introduction: Mass Spectrometry

Mass spectrometry (MS) is an analytical techniques used for separate charged ion based on their mass to charge ratio ( $m/z$ ). In proteomics and metabolomics, it is used to measure the  $m/z$  of peptide and metabolite ions. It is composed of:

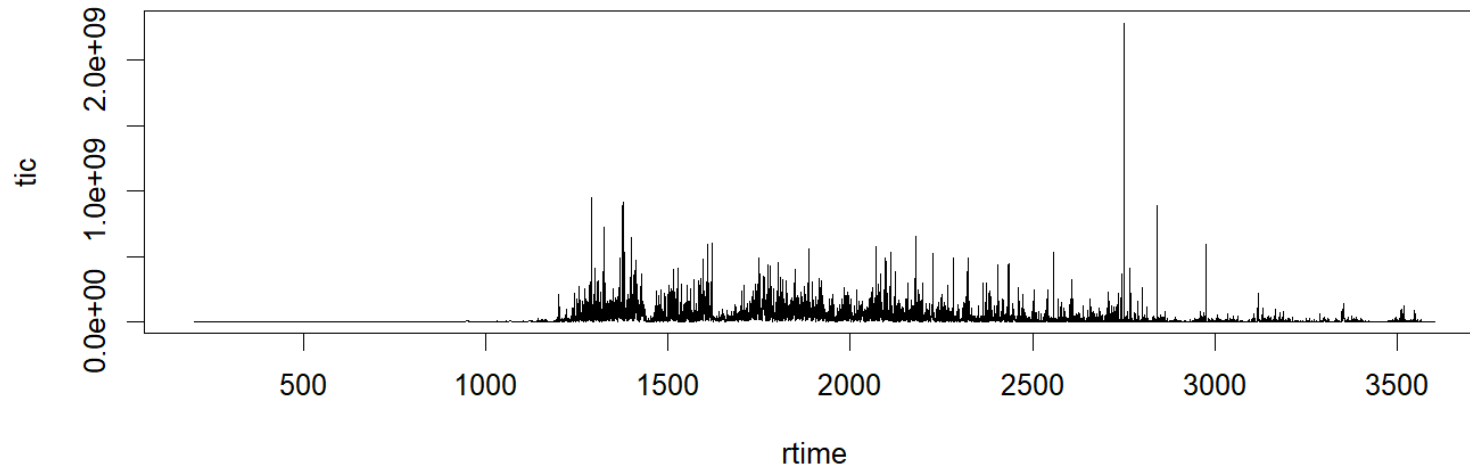
- Ion source: ionizes the molecules under analysis (analyte)
- Mass analyzer: separates the ions e.g time of flight (TOF), Orbitrap, quadrupole or ion traps
- Detector: quantifies the ions

MS is commonly coupled with liquid chromatography (LC-MS), thereby combining the physical separation abilities of LC with  $m/z$  analysis of MS. Also, gas chromatography can be coupled with MS (GC-MS).

**MS Level 1**



**MS Level 2**



## Resources/References

- Langfelder, P. and Horvath, S. (2008). WGCNA: an R package for weighted correlation network analysis. BMC Bioinformatics 9, 559.
- [https://edo98811.github.io/WGCNA\\_official\\_documentation/](https://edo98811.github.io/WGCNA_official_documentation/)
- <https://rformassspectrometry.github.io/book/>

